PHYSICS

NEWTON'S LAWS

British physicist Isaac Newton proposed a series of laws of motion, which he published in his masterpiece *Principia* (see 1698–99). Newton's laws describe how objects move, remain at rest, or interact with other objects and forces. Newton also formulated the law of universal gravity to describe the force of attraction that acts between physical bodies.

LAW	DESCRIPTION
First law of motion	Unless disturbed by a force, an object will either stay still or travel at a constant speed in a straight line.
Second law of motion	The force acting on a body is equal to its rate of change of momentum, which is the product of its mass and its acceleration.
Third law of motion	For every action there is an equal and opposite reaction.

LAW OF UNIVERSAL GRAVITATION

$$F = \frac{Gm_1m_2}{r^2}$$

F = force
G = universal gravitational constant
m_1, m_2 = masses
r^2 = square of distances between masses

MECHANICS OF FORCES

A force is a push or pull that makes an object move in a straight line or turn. Forces can act alone or together, and they can be harnessed to make machines work more effectively. The relationship between the properties of moving objects—such as time, distance, direction and speed—can be described using equations.

MOTION FORMULAE

QUANTITY	DESCRIPTION	FORMULA
Speed	$\frac{\text{distance}}{\text{time}}$	$S = \frac{d}{t}$
Time	$\frac{\text{distance}}{\text{speed}}$	$t = \frac{d}{S}$
Distance	speed × time	d = St
Velocity	$\frac{\text{displacement (distance in a given direction)}}{\text{time}}$	$v = \frac{s}{t}$
Acceleration	$\frac{\text{change in velocity}}{\text{time taken for change}}$	$a = \frac{(v-u)}{t}$
Resultant force	mass × acceleration	F = ma
Momentum	mass × velocity	p = mv

EQUATIONS OF MOTION UNDER CONSTANT ACCELERATION

These four equations are used to express constant acceleration in different ways:

$$s = \frac{(u + v)t}{2}$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$s = ut + \tfrac{1}{2}at^2$$

s = displacement
u = original velocity
v = final velocity
a = acceleration
t = time taken

HOOKE'S LAW

$$F_s = -kx$$

F_s = force of spring
k = spring constant (indication of spring's stiffness)
x = extension

TURNING FORCES

FORCE	DESCRIPTION	FORMULA	KEY
Moment of inertia	The equivalent of mass for an object rotating about an axis	$I = mr^2$	I = moment of inertia m = mass r^2 = square of distance from axis
Angular velocity	The velocity of an object rotating about an axis	$\omega = \frac{\Delta\theta}{\Delta t}$	ω = angular velocity Δθ = angular displacement Δt = change in time
Angular momentum	The momentum of an object rotating about an axis	L = Iω	L = angular momentum I = moment of inertia ω = angular velocity

LAWS OF THERMODYNAMICS

Thermodynamics is the study of the interrelationship between heat, work, and internal energy. The laws of thermodynamics describe what happens when a thermodynamic system goes though an energy change. Energy cannot be created or destroyed (as stated in the first law), but it can be converted into other forms.

LAW	DESCRIPTION
First law	Energy can be neither created nor destroyed.
Second law	The total entropy of an isolated system increases over time.
Third law	There is a theoretical minimum temperature at which the motion of the particles of matter would cease.
Zeroth law	If two bodies (distinct systems) are each in thermal equilibrium with a third body, these two bodies will also be in thermal equilibrium with each other.

TEMPERATURE SCALES

Heat is a form of kinetic energy. The amount of energy contained in an object—its temperature—can be measured using one of three scales: kelvin (one of the SI base units), Celsius (an SI derived scale), and Fahrenheit (first proposed by the Dutch–German–Polish physicist Daniel Gabriel Fahrenheit (see 1724). Absolute 0 kelvin is the point at which atoms in a substance have no heat energy and do not vibrate at all.

	KELVIN	CELSIUS	FAHRENHEIT
	373 K	100°C	212°F
	300 K	27°C	81°F
	273 K	0°C	32°F
	255 K	–18°C	0°F
	200 K	–73°C	–99°F
	100 K	–173°C	–279°F
Absolute zero	0 K	–273°C	–460°F